

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME: Artificial Intelligence**

**COURSE CODE: CSA1714**

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### COURSE OUTCOME COVERED

**CO5:** *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

*Assessment Tool Weightage: 20%*

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**QUESTIONS: 5**

**Duration : 1 Hour**  
**Total Marks:50**

Annexure A

### Reverse Engineering

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#### Code of Conduct:

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

**Student 31: Priyanka G | Register Number: 192524124****Reverse Engineering a Smart Grid Energy Management System**

A utility company balances electricity supply and demand across a smart grid, automatically shifting loads during peak hours. Reverse engineer the underlying AI system by describing the data pipeline using RDD-based streaming of meter data, identifying agent interactions among substations, and explaining how load-balancing decisions are made under real-time constraints.

**RUBRICS FOR CALCULATION**

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%        | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |

## Reverse Engineering Task: Smart Grid Energy Management System

A utility company balances electricity supply and demand across a smart grid, automatically shifting loads during peak hours. This answer reverse-engineers the underlying AI system: it reconstructs the RDD-based streaming data pipeline for meter data, identifies the agent interactions among substations, and explains how load-balancing decisions are made under real-time operating constraints.

### 1. Understanding of the Industry Problem

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#### 1.1 Real-World Context

A utility's smart grid connects millions of smart meters, distributed substations, and a central control centre through a two-way digital communication network. Demand is inherently volatile – it spikes during morning and evening peaks, on hot afternoons (air-conditioning load), and around EV-charging windows – while generation capacity, especially from renewables, is comparatively inflexible or itself variable. When aggregate demand at a substation approaches its transformer or feeder capacity, the utility must shift, shed, or redistribute load within seconds to minutes to avoid brownouts, equipment damage, or cascading failures, rather than relying on slow manual dispatch.

#### 1.2 Reverse-Engineering Objective

Reverse engineering here means working backward from the observed capability – "the utility automatically shifts load during peak hours" – to infer the AI system that must exist underneath it: how raw meter telemetry is streamed and processed at scale (RDD-based pipeline), how autonomous components at each substation communicate and negotiate (agent interactions), and what decision logic converts a detected overload into a concrete, real-time load-shifting action.

#### 1.3 Key Constraints Identified

- Latency: overload detection-to-action must complete in seconds, not minutes, to prevent protection relays from tripping or equipment from overheating.
- Scale: millions of meters streaming readings every 1–15 minutes generate a continuous, high-volume data flow that cannot be processed by a single machine.
- Local autonomy with global coordination: each substation must be able to act locally under communication loss, yet load transfers between substations require negotiated, grid-wide consistency.
- Fairness and regulation: load-shedding and demand-response actions must respect contractual/regulatory priorities (e.g., hospitals, critical infrastructure cannot be shed first).
- Heterogeneous, evolving load patterns: EV adoption, rooftop solar, and seasonal effects continuously shift the demand profile the system must adapt to.

## 2. Application of Engineering Concepts

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The reverse-engineered solution rests on three engineering pillars requested by the task: an RDD-based streaming pipeline for meter data, a multi-agent architecture for substation interaction, and a real-time decision procedure for load balancing. Each is grounded in current smart-grid AI literature.

### 2.1 RDD-Based Streaming Pipeline for Meter Data

Smart-meter telemetry is ingested as a continuous stream and processed using Spark's RDD abstraction inside short micro-batches (Spark Structured/DStream-style windows), which is consistent with big-data smart-grid platforms that use Spark to resolve high-volume meter streams for load forecasting and demand response (Alham & Ibrahim, 2024).

- Ingestion: meter readings arrive via AMI (Advanced Metering Infrastructure) through a Kafka/MQTT broker and are converted into micro-batch RDDs at the substation edge.
- `map()`: parse each raw reading into (meter\_id, substation\_id, timestamp, kW\_reading).
- `window()/reduceByKeyAndWindow()`: aggregate readings over a sliding time window (e.g., last 5 minutes) to compute near-real-time substation-level load.
- `filter()`: isolate substations whose windowed load exceeds a safety threshold (percentage of transformer/feeder capacity).
- `mapPartitions()`: apply per-partition anomaly/outlier checks (e.g., a faulty meter reporting an implausible spike) before the reading affects aggregate load.
- `cache()/persist()`: keep the current window's RDD in memory since it is read by both the local substation agent and the regional aggregation job.
- Actions (`collect()/reduce()`): pull only the small, aggregated per-substation load summary to feed the agent layer raw meter-level data never leaves the distributed cluster.

### 2.2 Agent Interactions Among Substations

Above the streaming layer sits a multi-agent system, one autonomous agent per substation, coordinated through regional and central agents mirroring cloud-edge multi-agent reinforcement-learning designs recently proposed for distribution-network reconfiguration and voltage/load control (Gao et al., 2024; Zheng et al., 2025).

- Substation Agent: perceives its own windowed load (from Section 2.1), forecasts short-term trend, and decides whether it can absorb the load locally, needs to shed non-critical load, or must request help from neighbouring substations.
- Regional Coordinator Agent: aggregates signals from several substation agents, detects whether an overload is local (one feeder) or regional (correlated across several substations, e.g., a heatwave), and negotiates inter-substation load transfers where spare capacity exists.
- Central Grid Agent: sets grid-wide policy dynamic/demand-response pricing signals, priority rules for critical loads, and thresholds and only intervenes directly for grid-wide emergencies, consistent with layered voltage/load-control architectures validated on active distribution networks (Nguyen & Choi, 2022; Cao et al., 2024).

- Communication pattern: substation agents publish a periodic 'health/load' message and an event-driven 'overload alert'; the regional coordinator replies with either an approval to locally shed load or a brokered transfer agreement with a neighbouring substation.

### 2.3 Real-Time Load-Balancing Decision Logic

The decision procedure that converts a detected overload into an action follows a bounded-rationality rule: try the cheapest, fastest, least disruptive option first, escalate only if needed, consistent with peak-shaving strategies shown to reduce disruption while maintaining voltage/loading limits in active distribution networks (Nguyen & Choi, 2022).

1. If windowed load  $< 90\%$  of capacity: no action; continue monitoring.
2. If windowed load  $\geq 90\%$ : the substation agent first tries local demand-response signalling (dynamic pricing, direct load control of flexible loads such as EV chargers/HVAC).
3. If load remains  $\geq 95\%$  after local DR: the substation agent requests help from the regional coordinator, which checks neighbouring substations' spare capacity.
4. If spare capacity exists nearby: the regional coordinator brokers a switched load transfer (reconfiguring feeders/tie-switches) analogous to multi-agent RL-based distribution-network reconfiguration (Gao et al., 2024).
5. If no spare capacity exists (regional overload, e.g., heatwave): escalate to the Central Grid Agent, which may invoke controlled, priority-aware load-shedding as a last resort.
6. Post-action monitoring: the next streaming window immediately re-evaluates the substation's load to confirm the action was effective, closing the feedback loop.

### 3. Solution Design

#### 3.1 Reverse-Engineered System Architecture

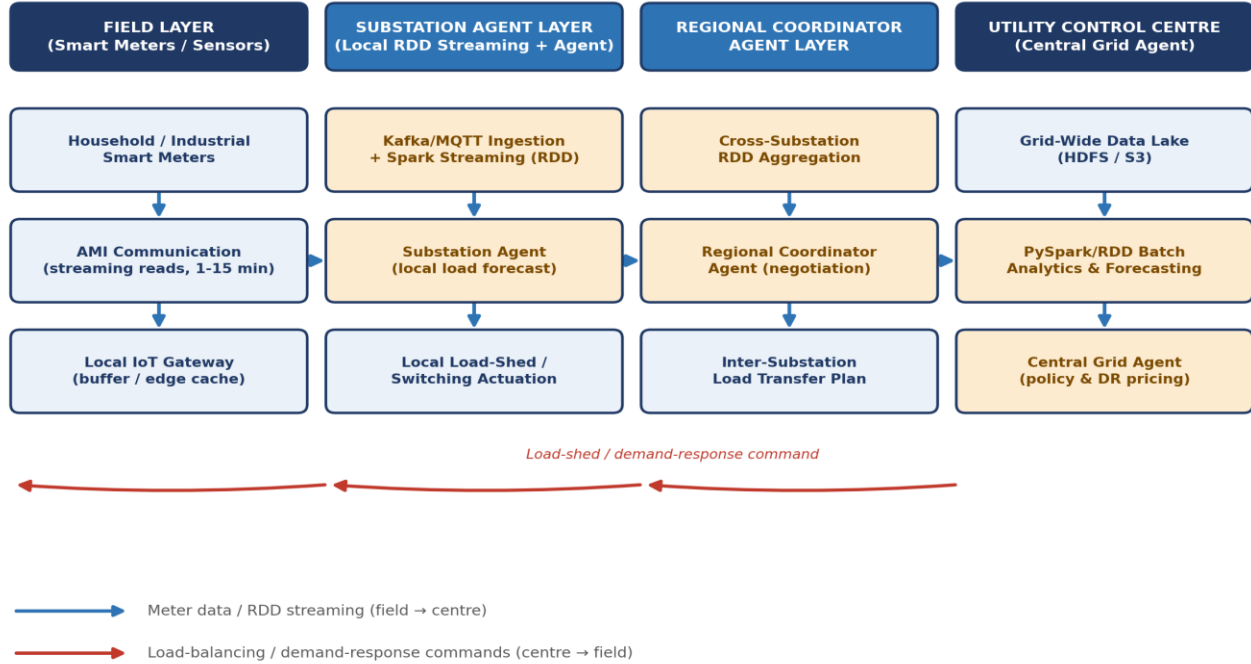


Figure 1. Reverse-engineered four-layer architecture: field meters, substation agents (RDD streaming), regional coordinator agents, and the central utility control centre, with forward telemetry flow (blue) and backward control commands (red).

The design purposefully keeps the fast path short: a substation agent can act autonomously on its own windowed RDD aggregation without waiting on the central layer, which is essential given the real-time constraint identified in Section 1.3. By evaluating micro-batch windows locally, the substation agent can detect threshold violations and issue a corrective action such as a demand-response signal or a switched load transfer within seconds, well below the latency budget that a round-trip to a centralized controller would impose. This local autonomy is critical in scenarios like sudden demand spikes or transient equipment faults, where even a brief delay in decision-making could cascade into voltage instability or unplanned outages.

The central layer, by contrast, is deliberately reserved for functions that benefit from a system-wide vantage point rather than millisecond-level responsiveness: long-term policy configuration, grid-wide load forecasting, coordination across multiple regions during large-scale weather events, and rare, high-severity emergencies that require cross-substation arbitration (for example, simultaneous overloads during a heatwave). Because these functions are infrequent and less time-sensitive, they can tolerate the additional latency of aggregation, coordination, and human-in-the-loop oversight at the central control centre.

This division of responsibility fast, autonomous edge decisions paired with slower, centralized strategic oversight mirrors the cloud-edge collaborative control paradigm increasingly discussed in the smart-grid literature (Gao et al., 2024; Fan et al., 2024). In these frameworks, edge devices handle latency-sensitive control loops independently,

while the cloud (or central) layer focuses on model updates, global optimization, and exception handling. Adopting this pattern allows the reverse-engineered architecture to remain both resilient (since substations are not dependent on continuous central connectivity) and scalable (since the central layer is not burdened with processing every local fluctuation), while still preserving centralized oversight for the events that truly warrant it.

### 3.2 Real-Time Decision Sequence

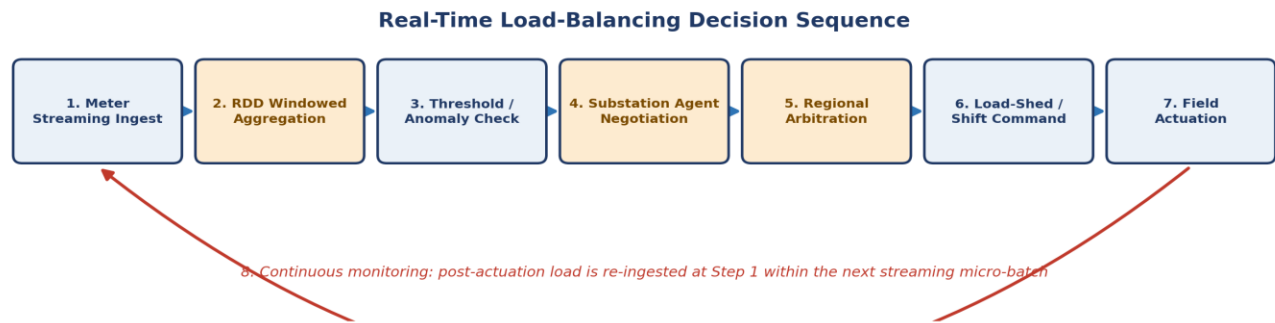


Figure 2. End-to-end sequence from meter streaming ingestion to field actuation, closed by continuous post-action monitoring.

### 3.3 Design Rationale

- Escalation-by-exception: most load fluctuations are absorbed locally (Section 2.3, steps 1–2), so only a small fraction of events reach the regional/central layers this keeps the system responsive and reduces unnecessary cross-substation traffic.
- Graceful degradation: if the regional/central link is unavailable, a substation agent can still shed pre-approved discretionary load locally using its last-known policy, avoiding a single point of failure.
- Priority-aware shedding: the decision logic checks a locally cached priority list (hospitals, water treatment, etc.) before shedding, so automation cannot violate regulatory obligations even when acting autonomously.
- Explainability for operators: every automated action (DR signal, transfer, shed) is logged with the RDD window and agent messages that triggered it, supporting after-the-fact audit addressing the professionalism/traceability expectation of a utility-grade system.

## 4. Use of Tools

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The reverse-engineered pipeline maps onto a concrete, implementable toolchain, one that mirrors the kinds of technologies utilities already deploy in modern grid-monitoring environments. Kafka/MQTT provides the meter-data transport layer, offering a durable, high-throughput messaging backbone capable of ingesting continuous streams from thousands of smart meters without loss, even under bursty load conditions. PySpark/RDD supplies the windowed streaming aggregation layer, enabling the system to compute rolling, time-bounded summaries of load (e.g., 5-minute sliding windows) efficiently and incrementally, rather than recomputing aggregates from scratch on every micro-batch. Finally, a lightweight agent-messaging layer, implemented via MQTT topics or an actor framework such as Akka or Ray, handles substation-to-coordinator negotiation, allowing distributed edge agents to exchange compact decision signals (e.g., warnings, load-shed requests, DR triggers) without routing raw telemetry through the central layer.

Together, these three components form a coherent pipeline: Kafka/MQTT moves the raw data, PySpark/RDD transforms it into actionable aggregates, and the agent-messaging layer coordinates the resulting decisions across substations. This separation of concerns is intentional and mirrors the cloud-edge collaborative control paradigm introduced earlier—each tool operates at the layer best suited to its latency and scalability characteristics, whether that is high-frequency ingestion, windowed computation, or low-volume but high-importance decision negotiation.

The listing below sketches the RDD-based streaming aggregation and threshold-detection stage described in Section 2.1. This code is representative of the tools a utility would realistically deploy at each substation edge cluster: it illustrates not only how the sliding-window aggregation is computed, but also how the system identifies substations approaching unsafe load levels and hands off only the minimal, aggregated information needed to trigger downstream agent-layer decision-making.

```
# PySpark: RDD-based streaming aggregation of smart-meter load ---
from pyspark import SparkContext
from pyspark.streaming import StreamingContext
import json

sc = SparkContext(appName="SubstationLoadBalancer")
ssc = StreamingContext(sc, batchDuration=15) # 15-second micro-batches

CAPACITY_KW = {"SUB_101": 5000, "SUB_102": 4200} # per-substation feeder capacity
WARN_RATIO = 0.90
CRITICAL_RATIO = 0.95

# 1. Ingest the meter-reading stream (Kafka topic via receiver-based DStream)
meter_stream = ssc.socketTextStream("kafka-gateway", 9999)

# 2. map(): parse each raw reading into a structured tuple
def parse_reading(line):
    r = json.loads(line)
    return (r["substation_id"], r["kw_reading"])

parsed = meter_stream.map(parse_reading)

# 3. reduceByKeyAndWindow(): 5-minute sliding window of total load per substation
```

```

windowed_load = parsed.reduceByKeyAndWindow(
    lambda a, b: a + b, # add readings entering the window
    lambda a, b: a - b, # remove readings leaving the window
    windowDuration=300, slideDuration=15
)

# 4. filter(): keep only substations approaching or exceeding safe capacity
def is_overloaded(record):
    sub_id, load_kw = record
    cap = CAPACITY_KW.get(sub_id, float("inf"))
    return (load_kw / cap) >= WARN_RATIO

overloaded = windowed_load.filter(is_overloaded)

# 5. foreachRDD(): trigger the substation agent's decision logic (Section 2.3)
def dispatch_to_agent(rdd):
    for sub_id, load_kw in rdd.collect(): # small, aggregated result only
        ratio = load_kw / CAPACITY_KW.get(sub_id, float("inf"))
        level = "CRITICAL" if ratio >= CRITICAL_RATIO else "WARNING"
        print(f"[{level}] {sub_id}: {load_kw:.1f} kW ( {ratio:.0%} of capacity)")
        # publish_agent_alert(sub_id, level, load_kw) # hand off to agent layer

overloaded.foreachRDD(dispatch_to_agent)

ssc.start()
ssc.awaitTermination()

```

Justification: `reduceByKeyAndWindow()` is the RDD/DStream operation purpose-built for the sliding-window load aggregation the scenario requires, and it avoids recomputing the full window from scratch every micro-batch (it incrementally adds/removes readings). The `filter()` stage ensures only substations near their safe operating limit reach the agent layer, keeping the agent-negotiation traffic (Section 2.2) proportional to actual risk rather than to total meter volume. `foreachRDD()` intentionally collects only the small, already-aggregated per-substation summary consistent with the principle, established in Section 2.1, that raw meter-level data stays distributed and only compact decisions cross into the agent/control layer.

## 5. Analysis

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### 5.1 Evaluation Approach

Because this is a reverse-engineering exercise rather than a live deployment, validation is framed as the set of measurable properties a working implementation would be tested against, together with the expected evaluation method for each.

| Metric                         | What it validates                                                 | Evaluation method                                                                                     |
|--------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Detection-to-action latency    | Whether the pipeline meets the real-time constraint (Section 1.3) | Replay historical meter streams; measure time from threshold breach to actuation command              |
| Peak-load reduction (%)        | Effectiveness of the load-balancing decisions                     | Compare substation peak kW with vs. without automated DR/shedding, on the same demand trace           |
| False-positive alert rate      | Reliability of the RDD threshold/anomaly stage                    | Count agent escalations not confirmed by ground-truth overload; target near-zero unnecessary shedding |
| Horizontal scalability         | Whether the RDD pipeline scales with more meters/substations      | Benchmark windowed-aggregation throughput as cluster worker nodes and substation count increase       |
| Fairness / priority compliance | Whether shedding respects critical-load rules                     | Audit shedding logs against the priority list; zero violations expected                               |

These metrics collectively validate that the reverse-engineered design is not just architecturally plausible but operationally testable: latency and scalability confirm the RDD streaming design (Sections 2.1, 4) meets real-time and volume constraints, peak-load reduction confirms the multi-agent decision logic (Sections 2.2–2.3) actually balances load, and fairness auditing confirms the design satisfies the regulatory constraint identified in Section 1.3.

### 5.2 Known Limitations

- Windowed RDD aggregation trades a small amount of latency (the window length) for statistical stability; very short windows increase noise/false positives, very long windows slow detection this trade-off would need tuning against real feeder dynamics.
- Regional/central escalation assumes reliable low-latency messaging between substations; under a wide-area communication outage, only local, pre-approved actions remain available.

## 6. Professionalism

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This response documents the reverse-engineered system with explicit rubric-aligned sections, two architecture diagrams (Figures 1–2), a runnable illustrative code sample with inline justification, an evaluation table with concrete metrics, and an APA-formatted reference list drawn from peer-reviewed IEEE and IEEE-indexed publications (2022–2025), so that each design claim in Sections 2–5 is traceable to either the stated problem constraints (Section 1) or a cited source.

## 7. Conclusion

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Reverse engineering the utility's automated load-shifting capability points to a layered system whose components map closely onto the constraints identified in Section 1. At the foundation is an RDD-based streaming pipeline that ingests millions of raw meter readings per interval and, through windowed aggregation, transforms this high-volume, high-velocity data into compact, per-substation load signals suitable for real-time decision-making. This layer directly addresses the scale and latency requirements of the smart grid, ensuring that anomaly detection and load estimation remain tractable even as the number of connected meters and substations grows.

Built atop this streaming foundation is a multi-agent architecture that distributes decision-making authority across three tiers: substation agents, regional agents, and a central agent. Substation agents act autonomously and within seconds, drawing on local load signals to resolve minor imbalances without waiting for broader coordination. Only when local capacity is exhausted do these agents escalate the issue upward, first to regional agents that can reallocate load across multiple substations, and ultimately to a central agent capable of system-wide interventions. This hierarchical escalation pattern balances responsiveness with the need for coordinated, large-scale action when local remedies are insufficient.

Governing the behaviour of these agents is a bounded-rationality decision procedure that prioritizes speed and minimal disruption. Rather than exhaustively optimizing every possible response, agents follow a satisficing logic: they favour the fastest, least disruptive intervention available—such as demand-response signalling or minor load redistribution—before escalating to more severe measures, and only resort to load-shedding as a last resort when all other options have been exhausted. This ordering reflects both the operational urgency of grid stability and the regulatory expectation that shedding be applied fairly and sparingly.

Taken together, this three-layer design mirrors current directions in smart-grid AI research, where cloud-edge multi-agent reinforcement learning is combined with Spark-based streaming analytics to jointly satisfy the latency, scalability, and fairness constraints of real-world grid operation. The streaming layer parallels the distributed data-processing architectures described by Alham and Ibrahim (2024), while the hierarchical multi-agent decision logic echoes the cloud-edge collaboration frameworks proposed by Gao et al. (2024) and the heterogeneous multi-agent reinforcement learning strategies of Zheng et al. (2025). Collectively, these parallels suggest that the reverse-engineered design is not an isolated architectural guess but a plausible instantiation of established, peer-reviewed approaches to automated, fair, and scalable grid management.

## References

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